



Advanced Petroleum Economics and Decision Analysis

Tools for Uncertainty Assessment, valuation, and Investment Strategy



Advanced Petroleum Economics and Decision Analysis

The energy economy is transforming faster than any industry in history. Success in this new era of environment, competition, and emerging technologies depends not on chance, but on informed, strategic decisions. As Peter Drucker once noted, “The greatest danger in times of turbulence is not the turbulence—it is to act with yesterday’s logic.” This course provides advanced tools to think clearly energy business and challenges ahead—using data, economic reasoning, and decision analysis—to turn global challenges into sustainable opportunities for growth, resilience, and innovation.

Participants will master probabilistic modelling, real options valuation, and Bayesian reasoning—tools essential for evaluating complex upstream projects under uncertainty. The curriculum integrates financial theory, policy economics, and scenario-based planning to support informed decision-making.

Software Requirements: The course uses **Lumivero’s DecisionSuit Tools** (formerly Palisade DecisionSuit Tools), including @risk and PrecisionTree. However, delegates do not need access to these platforms as other alternative solutions will be discussed during the course.



LOCATION

Dubai, UAE

Houston, USA

London, UK

Kuala Lumpur, Malaysia

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



COURSE OBJECTIVES

This course highlights:

- Apply probabilistic valuation techniques to upstream oil and gas assets.
- Construct and interpret Monte Carlo simulations for reserves, price, and cost uncertainty.
- Use Bayesian updating to refine probability estimates as new information emerges.
- Evaluate real options embedded in petroleum investment decisions.
- Model probabilistic price scenarios using stochastic processes and market data.
- Integrate fiscal regimes, ESG constraints, and policy uncertainty into investment appraisals.
- Develop decision support frameworks that combine technical, economic, and strategic dimensions.

To succeed in the oil & gas industry is to create value. Learning about the subsurface challenges, along with the energy commodity market leads to insights that support good decision making. Good project decisions often lead to value creation.



At the end of this training course, you will learn to:

- Explain key aspects of the evolving energy landscape and its implications for oil and gas ventures.
- Summarise the fundamental principles of petroleum project economics and valuation techniques.
- Apply prudent business decisions to differentiate successful petroleum firms in an evolving global landscape.
- Analyse the impact of risk and uncertainty on management decision-making.
- Apply consistent financial valuation techniques to analyse petroleum projects.
- Use analytical tools and software for effective project evaluation and decision making.
- Evaluate the implications of uncertainty and risk on project feasibility
- Create decision support models that consider the influence of uncertain hydrocarbon prices and environmental considerations on value creation

Training Methodology

- Hands-on exercises using financial and statistical models.
- Real-world case studies covering successful and failed projects.
- Expert-led discussions on policy impacts and strategic decision-making.
- Collaborative scenario planning to develop strategic adaptability.
- Software-based simulation for uncertainty quantification.

Day 1

Strategic framing and decision architecture

- Global energy landscape and geopolitical factors.
- Decision quality and framing in high-capital energy projects
- Structuring complex decisions: influence diagrams and decision trees
- Value chain: economic principles underlying petroleum decision-making.
- Systematic approaches to managing uncertainty.
- Introduction to Bayesian decision theory
- **Case Study:** framing a multi-stage offshore field development project.

Day 2

Probabilistic Reserves & Uncertainty Quantification

- Market-based approach to investment decision making
- Reserve classification: deterministic vs probabilistic approaches
- Building probabilistic reserve models using P10/P50/P90 distributions
- Bayesian updating for reserve estimates with new seismic or drilling data
- Monte Carlo simulation for volumetric uncertainty
- **Case Study:** Probabilistic valuation of a deepwater prospect with limited data

Day 3

Uncertainty Analysis & Quantitative Methods

- Advanced Discussions on Investment decision techniques: NPV, IRR, and Payback period.
- Sensitivity analysis in economics analysis of projects.
- Monte Carlo fundamentals: distributions, iterations, convergence
- Modelling price, cost, and schedule uncertainty in upstream projects
- Correlation structures and joint probability modelling
- Using @Risk and PrecisionTree for integrated project simulation
- **Case Study:** Full-cycle economic model of Enceladus field with probabilistic inputs.



Day 4

Strategic Planning & Policy Economics

- Economic impact of energy transition policies.
- Real options theory: defer, expand, abandon, switch
- Valuation techniques: binomial trees, Black-Scholes, decision trees
- Strategic value of flexibility under volatile price and policy regimes
- Linking real options to corporate portfolio strategy
- **Case:** Valuing the option to delay Enceladus development amid carbon pricing uncertainty.

Day 5

Practical Applications & Project Presentations

- Modelling commodity prices: mean reversion, GBM, jump diffusion
- Building stochastic price scenarios for long-term planning
- Sensitivity analysis of investment decisions
- Scenario planning with policy, technology, and market disruptions
- Future trends: carbon markets, ESG considerations, and sustainable energy investments.
- Capstone: Presenting a probabilistic investment recommendation for Enceladus field





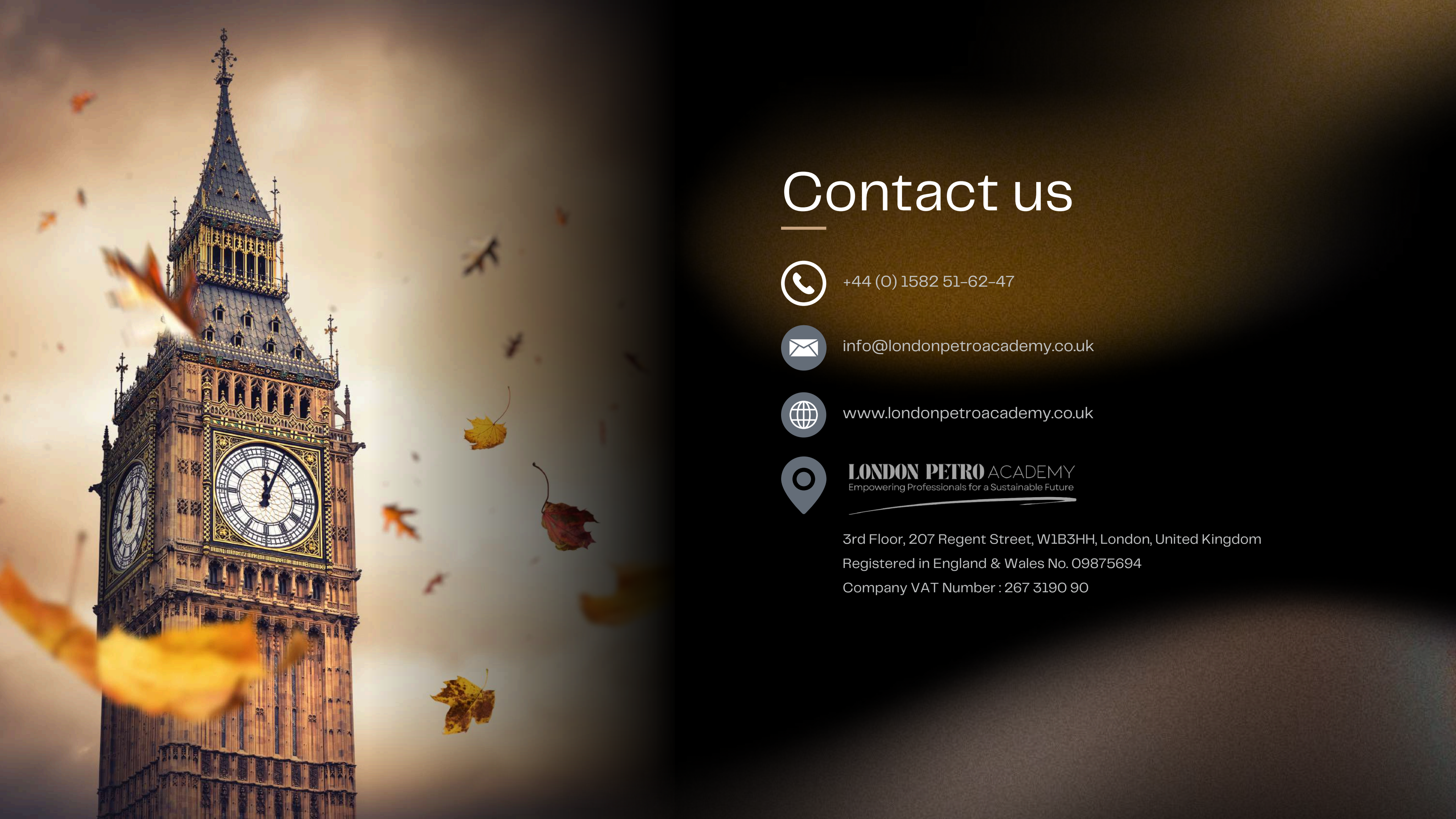
BABAK JAFARIZADEH

Dr. Babak Jafarizadeh is a senior consultant and engineering economist who views engineering as the pursuit of economic solutions to technical challenges in an uncertain world. With extensive experience in the oil and gas industry, he has served as a senior economic analyst supporting multi-million-dollar investment decisions for projects across the globe.

Since 2019, Dr. Jafarizadeh has worked as a senior consultant for leading training and consulting companies, while also holding academic appointments at several prestigious institutions — including Heriot-Watt University in the UK (Assistant Professor), the University of California, Berkeley (Lecturer at the Fung Institute for Engineering Leadership), and previously the University of Greenwich (Senior Lecturer). His earlier industry experience includes serving as a Senior Economic Analyst with Equinor (formerly Statoil) in Norway and the USA.

Dr. Jafarizadeh holds a PhD in Petroleum Engineering with a concentration in Investment and Decision Analysis from the University of Stavanger (in collaboration with the University of Texas at Austin), as well as an MSc in Financial Engineering and a BSc in Industrial Engineering. He is an active member of several professional associations, including the Society of Petroleum Engineers (SPE), the Institute for Operations Research and the Management Sciences (INFORMS), and the European Association of Geoscientists and Engineers (EAGE).

His teaching and research interests lie at the intersection of decision analysis and corporate finance, with particular applications in engineering and the energy industry. He has designed and delivered numerous professional training courses, such as Advanced Financial Modelling for Oil and Gas Professionals, Advanced Project Economics, Oil and Gas: Markets, Trading, Pricing and Economic Frameworks, and Energy Economics. Dr. Jafarizadeh is also the author of the book Economic Decision Analysis (for Project Feasibility Studies) and continues to contribute to both academia and industry through research, consulting, and executive education. His areas of expertise include petroleum economics, investment and decision analysis, project feasibility studies, corporate finance, and financial modelling for energy projects.



Contact us



+44 (0) 1582 51-62-47



info@londonpetroacademy.co.uk



www.londonpetroacademy.co.uk



LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future

3rd Floor, 207 Regent Street, W1B3HH, London, United Kingdom

Registered in England & Wales No. 09875694

Company VAT Number : 267 3190 90